Integrated Academy of Management & Technology

Delhi-Hapur Bypass (NH-9 near Dasna Flyover Udyog Kunj,
Ghaziabad, Uttar Pradesh 201015

IT Department

Subject Syllabus and University Evaluation Scheme

BCA 305P Minor Project BCA V SEM

Faculty Teaching in Different section is:
Mr. Kamal Nain

Code: BCA-305P
Course Objectives **Teaching Hours: 40**

- **Understand** the fundamental concepts, characteristics, types, and life cycle of software projects and Software Development Life Cycle (SDLC) models.
- **Analyze** project problems, identify stakeholder requirements, perform requirement analysis, and prepare Software Requirements Specification (SRS) and feasibility reports.
- **Apply** software design techniques, UML modeling, system architecture, and database design principles to develop structured software solutions.
- **Develop** functional software applications using appropriate frontend, backend, database, and integration technologies.
- **Apply** software testing, debugging, deployment, maintenance, and documentation techniques to successfully complete and present a software project.

Syllabus

Unit I – Introduction to Software Projects and SDLC

Introduction to Software Projects, meaning and characteristics of software projects, objectives of software projects, types of software projects, software project stakeholders, software project team, roles and responsibilities, project life cycle, software development process, Software Development Life Cycle (SDLC), importance of SDLC, phases of SDLC, Waterfall Model, Incremental Model, Iterative Model, Prototype Model, Spiral Model, Agile Model, comparison of SDLC models, selection of appropriate SDLC model, project planning and project roadmap.

Unit II – Requirement Analysis and Feasibility Study

Introduction to Requirement Engineering, problem identification, problem definition, project objectives, stakeholder identification, requirement gathering, requirement elicitation techniques, functional requirements, non-functional requirements, user requirements, system requirements, requirement analysis, requirement prioritization, requirement validation, Software Requirements Specification (SRS), characteristics of a good SRS, SRS structure, feasibility study, technical feasibility, economic feasibility, operational feasibility, schedule feasibility, legal feasibility, resource feasibility, feasibility report, project scope, assumptions, constraints, risks, project proposal and project approval.

Unit III – Software Design, UML and Database Design

Introduction to software design, design objectives, design principles, system architecture, modular design, Unified Modeling Language (UML), importance of UML, Use Case Diagram, Class Diagram, Activity Diagram, Sequence Diagram, Component Diagram, Deployment Diagram, actors and use cases, UML relationships, database fundamentals, database design process, entities, attributes, relationships, Entity Relationship (ER) Diagram, primary key, foreign key, candidate key, normalization, database tables, relational database design, SQL basics, database connectivity and database security considerations.

Unit IV – Frontend and Backend Development

Introduction to frontend development, User Interface (UI) and User Experience (UX), HTML, CSS, JavaScript, responsive web design, forms, navigation, form validation, Bootstrap, frontend project structure, introduction to backend development, client-server architecture, server-side programming, backend languages and frameworks, database connectivity, CRUD operations, authentication, authorization, sessions, APIs, form processing, error handling, input validation, security basics, frontend-backend integration and project implementation.

Unit V – Testing, Deployment and Documentation

Introduction to software testing, importance of testing, testing principles, testing process, test planning, test cases, test scenarios, Unit Testing, Integration Testing, System Testing, Acceptance Testing, Black-box Testing, White-box Testing, Functional Testing, Non-functional Testing, Regression Testing, Usability Testing, debugging, defect reporting, test report, software deployment, local deployment, web hosting, server environment, domain and hosting concepts, database deployment, application configuration, version control, Git and GitHub, application maintenance, project documentation, project report structure, user manual, technical documentation, installation guide, screenshots, project presentation, viva preparation, project evaluation and final submission.

Course Outcomes:

Course Outcomes

S. No.	Course Outcome	PO Attainment
CO1	**Understand** the fundamentals of software projects, project life cycle, SDLC phases, and different software development models.	**PO1**
CO2	**Analyze** project problems, stakeholder requirements, functional and non-functional requirements, and prepare Software Requirements Specification (SRS) and feasibility reports.	**PO2**
CO3	**Apply** software design principles, UML modeling techniques, system architecture, and database design concepts for software projects.	**PO3**
CO4	**Develop** frontend and backend components and integrate software applications with databases using appropriate development technologies.	**PO3**
CO5	**Apply** software testing, debugging, deployment, and maintenance techniques to develop reliable and functional software applications.	**PO4**
CO6	**Create** complete software project documentation, user manuals, presentations, and demonstrate the developed project through project presentation and viva.	**PO5**

Scope of Subject

Software Project Development provides students with practical knowledge of the complete software development process, from problem identification to final deployment and documentation. The subject covers project planning, SDLC models, requirement analysis, feasibility study, SRS preparation, UML modeling, database design, frontend and backend development, testing, debugging, deployment, and maintenance. It enables students to convert real-world problems into functional software solutions using appropriate technologies and development methodologies. Through practical activities and project-based learning, students develop skills in system analysis, software design, programming, database management, testing, documentation, and project presentation. The subject prepares students for careers in software development, web development, software testing, database development, system analysis, and project management, while providing a strong foundation for advanced studies and professional software project development.

Integrated Academy of Management & Technology

Delhi-Hapur Bypass (NH-9 near Dasna Flyover Udyog Kunj,
Ghaziabad, Uttar Pradesh 201015

IT Department

Lesson Plan
Minor Project BCA V SEM
Course Code: BCA 305P
COURSE OBJECTIVES:
➢ **Understand** the fundamental concepts, characteristics, types, and life cycle of software projects and Software Development Life Cycle (SDLC) models.
➢ **Analyze** project problems, identify stakeholder requirements, perform requirement analysis, and prepare Software Requirements Specification (SRS) and feasibility reports.
➢ **Apply** software design techniques, UML modeling, system architecture, and database design principles to develop structured software solutions.
➢ **Develop** functional software applications using appropriate frontend, backend, database, and integration technologies.
➢ **Apply** software testing, debugging, deployment, maintenance, and documentation techniques to successfully complete and present a software project.
COURSE OUTCOME:
Understand the basics of data analytics and its applications.
Develop proficiency in using spreadsheet software for data manipulation and analysis.
Apply software design principles, UML modeling techniques, system architecture, and database design concepts for software projects.
Develop frontend and backend components and integrate software applications with databases using appropriate development technologies.
Apply software testing, debugging, deployment, and maintenance techniques to develop reliable and functional software applications.
Build and use spreadsheet models for decision making and communicate data insights effectively.

Class	BCA V Semester	1. White Board & Marker
Section	A+B+C	2. PPT
Course Name	Minor Project	3. Audio-Visual
Course Code	BCA 305P	4. Program on Computer System
Faculty	Mr. Kamal Nain	5. Hands-on Practical Activity

S.no	Day	Topic/Presentation /Activity	Delivery Method
		Unit-I: Introduction to Software Projects & SDLC	
1	D1	Introduction to Software Projects, objectives and characteristics of software projects (L1)	PPT + Discussion
2	D2	Types of Software Projects and Project Stakeholders (L2)	PPT + Examples
3	D3	Software Project Team, Roles and Responsibilities (L3)	PPT + Discussion
4	D4	Software Project Life Cycle and Project Development Process (L4)	White Board + PPT

5	D5	Introduction to SDLC and SDLC Phases (L5)	PPT + Diagram
6	D6	Waterfall, Incremental and Iterative Models (L6)	PPT + Comparison
7	D7	Prototype, Spiral and Agile Models (L7)	PPT + Case Study
8	D8	Selection of SDLC Model and Project Roadmap (L8)	Discussion + Activity
Unit-II: Requirement Analysis & Feasibility Study			
9	D9	Introduction to Requirement Analysis and Problem Identification (L9)	PPT + Discussion
10	D10	Stakeholder Identification and Requirement Gathering Techniques (L10)	PPT + Activity
11	D11	Functional and Non-Functional Requirements (L11)	PPT + Examples
12	D12	User Requirements, System Requirements and Requirement Prioritization (L12)	White Board + PPT
13	D13	Software Requirements Specification (SRS): Structure and Contents (L13)	PPT + Sample SRS
14	D14	Feasibility Study: Technical, Economic and Operational Feasibility (L14)	PPT + Case Study
15	D15	Schedule, Legal and Resource Feasibility; Project Scope and Constraints (L15)	PPT + Discussion
16	D16	Project Proposal, SRS & Feasibility Report Workshop (L16)	Hands-on Activity
Unit-III: UML & Database Design			
17	D17	Introduction to Software Design, System Architecture and Design Principles (L17)	PPT + Diagram
18	D18	Introduction to UML, Actors, Use Cases and Relationships (L18)	PPT + White Board
19	D19	Use Case Diagram – Concepts and Project Implementation (L19)	Demo + Hands-on
20	D20	Class Diagram – Classes, Attributes, Methods and Relationships (L20)	Demo + Hands-on
21	D21	Activity Diagram and Sequence Diagram (L21)	White Board + Demo
22	D22	Component Diagram, Deployment Diagram and UML Review (L22)	PPT + Hands-on
23	D23	Database Fundamentals, DBMS, Tables, Records and Fields (L23)	PPT + Demo
24	D24	ER Diagram, Keys, Relationships and Database Design (L24)	White Board + Hands-on
Unit-IV: Frontend & Backend Development			
25	D25	Introduction to Frontend Development, UI/UX and Project Interface Design (L25)	PPT + Demo
26	D26	HTML: Page Structure, Forms, Tables and Navigation (L26)	Computer Practical
27	D27	CSS, Responsive Design and Bootstrap (L27)	Computer Practical
28	D28	JavaScript, Events and Form Validation (L28)	Computer Practical
29	D29	Introduction to Backend Development and Client-Server Architecture (L29)	PPT + Demo

30	D30	Backend Programming, Database Connectivity and CRUD Operations (L30)	Computer Practical
31	D31	Authentication, Authorization, Sessions and Input Validation (L31)	Demo + Practical
32	D32	Frontend-Backend Integration and Project Implementation (L32)	Hands-on Activity
Unit-V: Testing, Deployment & Documentation			
33	D33	Introduction to Software Testing, Testing Process and Test Planning (L33)	PPT + Discussion
34	D34	Test Cases, Unit Testing, Integration Testing and System Testing (L34)	PPT + Practical
35	D35	Black-Box, White-Box, Functional and Non-Functional Testing (L35)	PPT + Examples
36	D36	Debugging, Defect Reporting and Testing Report Preparation (L36)	Hands-on Activity
37	D37	Software Deployment, Local Server, Hosting, Domain and Database Deployment (L37)	PPT + Demo
38	D38	Git, GitHub, Project Maintenance and Version Control (L38)	Computer Practical
39	D39	Project Documentation: Report, User Manual, Screenshots and PPT (L39)	Workshop
40	D40	Final Project Demonstration, Presentation, Viva and Course Review (L40)	Project Demo + Viva

Suggested Readings

Suggested Readings

Text Books

1. **Roger S. Pressman & Bruce R. Maxim**, *Software Engineering: A Practitioner's Approach*, 8th Edition, McGraw-Hill Education.
2. **Ian Sommerville**, *Software Engineering*, 10th Edition, Pearson Education.
3. **Rajib Mall**, *Fundamentals of Software Engineering*, 5th Edition, PHI Learning.
4. **Pankaj Jalote**, *An Integrated Approach to Software Engineering*, 3rd Edition, Springer.
5. **Craig Larman**, *Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development*, 3rd Edition, Pearson.

Reference Books

1. **Carlo Ghezzi, Mehdi Jazayeri & Dino Mandrioli**, *Fundamentals of Software Engineering*, 2nd Edition, Pearson Education.
2. **K. K. Aggarwal & Yogesh Singh**, *Software Engineering*, New Age International Publishers.
3. **Robert C. Martin**, *Clean Code: A Handbook of Agile Software Craftsmanship*, Prentice Hall.
4. **Robert J. Kauffman & James C. R.**, *Software Project Management*, relevant academic reference material.
5. **Martin Fowler**, *UML Distilled: A Brief Guide to the Standard Object Modeling Language*, 3rd Edition, Addison-Wesley.

Web Resources

1. **GeeksforGeeks – Software Engineering**
   https://www.geeksforgeeks.org/software-engineering/
2. **TutorialsPoint – Software Engineering**
   https://www.tutorialspoint.com/software_engineering/
3. **NPTEL – Computer Science and Engineering Courses**
   https://nptel.ac.in/
4. **IEEE Computer Society**
   https://www.computer.org/
5. **GitHub Documentation**
   https://docs.github.com/
6. **MDN Web Docs – Web Development**
   https://developer.mozilla.org/
7. **W3Schools – Web Development Tutorials**
   https://www.w3schools.com/
8. **MySQL Documentation**
   https://dev.mysql.com/doc/